

KUMARASAMY K

Senior Software Engineer – Android | Senior Android Developer
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SUMMARY

Senior Android Developer with **12+ years of software engineering experience** building, maintaining, and evolving Android applications across recruitment, HR, enterprise automation, banking, and device-integrated solutions. Strong expertise in **Kotlin, Java, Android SDK, Jetpack, Jetpack Compose, MVVM, Coroutines, Flow, Hilt, Dagger, Room, DataStore, Retrofit, and REST APIs**.

Experienced in taking ownership of complex application architecture, modernizing legacy Android codebases, resolving production-critical issues, improving application reliability, and mentoring engineers. Led an architectural migration involving **80+ Activities**, moving legacy persistence and dependency-management patterns toward Hilt and Jetpack DataStore while redesigning asynchronous data access using lifecycle-aware Coroutines.

Strong hands-on experience with **offline-first application development, BLE communication, external hardware and peripheral integration**, including thermal printers, barcode scanners, POS/card readers, biometric devices, and portable receipt printers. Experienced in diagnosing issues across Android, backend services, databases, stored procedures, and reporting layers.

Additional full-stack experience with **Node.js, Express, MySQL, Python, Flask, and FastAPI**, enabling effective collaboration and troubleshooting across client and backend systems. Experienced in **JUnit, Espresso, Firebase Crashlytics, LeakCanary, CI/CD, Jenkins, GitHub Actions, accessibility, TalkBack, localization, and WCAG-compliant Android development**.

KEY ENGINEERING STRENGTHS

Android Architecture & Modernization: Legacy Android application modernization; architecture migration and technical-debt reduction; MVVM and lifecycle-aware application design; dependency injection with Hilt/Dagger; reactive/asynchronous programming with Coroutines and Flow.

Production Debugging & Reliability: Root-cause analysis across Android, backend, SQL, and reporting layers; memory-leak investigation using LeakCanary and WeakReference; production crash investigation using Crashlytics; data-integrity issue diagnosis and remediation; zero-downtime data cleanup and corrective migrations.

Device & Connectivity Engineering: BLE communication; Bluetooth thermal printers; barcode scanners; biometric devices; POS/card readers; portable receipt printers; low-connectivity application environments.

Engineering Leadership: Technical mentoring; code reviews; architecture discussions; cross-layer troubleshooting; Agile development and sprint planning; production support and incident resolution.

TECHNICAL SKILLS

Programming Languages: Kotlin, Java, JavaScript, Python

Android Development: Android SDK, Jetpack, Jetpack Compose, MVVM, ViewModel, LiveData, Data Binding, Navigation Component, WorkManager

Architecture & Design: Application Architecture, MVVM, Dependency Injection, Hilt, Dagger, Legacy Application Modernization, State Management, Offline-First Architecture

Asynchronous Programming: Kotlin Coroutines, Flow, lifecycle-aware asynchronous processing

Data & Persistence: Room, SQLite, Jetpack DataStore, SharedPreferences, JSON, local caching, offline persistence

Networking & APIs: Retrofit, REST APIs, Moshi, Gson, JSON, API integration, asynchronous network communication

Hardware & Connectivity: Bluetooth Low Energy (BLE), Bluetooth peripherals, thermal printers, barcode scanners, POS/card readers, biometric devices, portable receipt printers

Backend Development: Node.js, Express, MySQL, Stored Procedures, Python, Flask, FastAPI, backend API development

Testing & Quality: JUnit, Espresso, Unit Testing, UI Testing, Crashlytics, LeakCanary, debugging, production issue analysis

DevOps & Tools: Git, GitHub, GitHub Actions, Jenkins, CI/CD, Android Studio, Firebase, Postman

Accessibility: TalkBack, screen-reader support, content labeling, focus navigation, scalable fonts, localization, WCAG-compliant Android development

Engineering Practices: Code Reviews, Agile/Scrum, Sprint Planning, Technical Mentoring, Production Support, Architecture Reviews

PROFESSIONAL EXPERIENCE

Senior Software Engineer – Android | Oct 2020 – Present

TalentMicro Innovations Pvt Ltd

- Lead Android application development and contribute to architectural decisions across Android and backend systems, working on complex enterprise workflows and production-critical functionality.

- Led an architectural migration of **80+ Activities** from legacy SharedPreferences-based persistence and dependency patterns to **Hilt dependency injection and Jetpack DataStore**, redesigning suspend-based data access across cached fields, lifecycle-scoped Coroutines, and legacy call sites.
- Consolidated fragmented ViewModel responsibilities into a centralized **TicketViewModel**, eliminating a redundant MainActivityViewModel, duplicate API call paths, and unnecessary state-management complexity.
- Integrated REST APIs using **Retrofit, Kotlin Coroutines, and Flow**, implementing asynchronous data handling and offline-first persistence using **SQLite/Room**.
- Diagnosed and resolved a static-reference memory leak in the login flow using **WeakReference**, preventing Activity instances from being retained across application sessions and addressing a potential source of production OOM crashes.
- Investigated and resolved a production SQL defect involving **COALESCE and LEFT JOIN** default handling that was silently mass-unpublishing live job postings. Reworked expiry logic with explicit null handling and introduced schema-level safeguards to prevent recurrence.
- Designed and executed a two-phase data-integrity remediation for duplicate records in a core organization-data table, combining an upsert-level prevention mechanism with a six-step cursor-based cleanup of corrupted records without service downtime.
- Diagnosed and resolved a multi-layer leave-approval workflow defect spanning **MySQL stored procedures and reporting views**, restoring correct approval-status visibility across the HR platform.
- Built and maintained **Node.js/Express backend services**, including migration of complex MySQL stored-procedure logic to parameterized Node.js queries supporting core job-matching workflows.
- Implemented Android accessibility improvements including **TalkBack support, meaningful labels, focus navigation, scalable UI elements, and screen-reader support**.
- Investigated application stability issues using **Firebase Crashlytics and LeakCanary**, identifying production defects and memory-related issues for corrective action.
- Participated in code reviews and provided architectural guidance to junior engineers, helping improve implementation quality and consistency across the development team.
- Collaborated across Android, backend, database, and reporting layers to diagnose production issues that could not be resolved through client-side investigation alone.
- Contributed to Agile development activities including technical discussions, sprint planning, implementation, debugging, code reviews, and production support.

Software Engineer – Android | Apr 2016 – Oct 2020

HireCraft Software Pvt Ltd

- Developed and maintained Android applications using **Kotlin and Java**, implementing application features across multiple product workflows.
- Built modular Android components using **MVVM architecture**, separating UI, presentation, and data responsibilities for maintainable application development.
- Integrated third-party SDKs and libraries to extend application capabilities and support product requirements.
- Worked on application performance improvements, debugging, and issue resolution across Android components.
- Implemented accessibility improvements including **localization, scalable fonts, and screen-reader support**.
- Participated in peer code reviews and contributed to development standards and code-quality improvements.
- Collaborated with cross-functional team members during sprint planning, implementation, testing, defect resolution, and Agile ceremonies.

Junior Android Developer | Jul 2014 – Apr 2016

Benchmark Business Software

- Developed and debugged core Android application features with a focus on application stability and performance.
- Implemented localization and accessibility functionality to improve usability across different users and environments.
- Designed and implemented **BLE communication** for a KOT (Kitchen Order Ticket) application.
- Integrated **Bluetooth thermal printers and barcode scanners** to enable real-time order-ticket printing at restaurant kitchen stations.
- Built BLE-based device communication for a rural banking collection application.
- Integrated **handheld biometric devices and portable receipt printers** to support field agents performing cash collection and passbook-printing operations.
- Worked with hardware communication in **low-connectivity rural environments**, requiring reliable device interaction and data handling.
- Investigated device and application issues during development and supported debugging of Android and peripheral-integration problems.

SELECTED PROJECTS

WhatMate | 2020 – Present | AI-Driven Enterprise Automation Platform | Android & Backend

- Developed Android application features supporting enterprise automation workflows.
- Integrated Android application functionality with backend services and APIs to support end-to-end task execution.
- Optimized client-to-backend communication to improve task execution efficiency and application responsiveness.

- Investigated and resolved integration issues across Android and backend layers.

• **10,000+ downloads | 4.1★**

Tallite | 2021 – Present | Global Recruitment Platform | Backend & Full-Stack Engineering

- Developed backend modules using **Flask and FastAPI** supporting recruitment workflows and job-management functionality.
 - Implemented services for job workflows, referral tracking, authentication, and real-time status updates.
 - Designed API-driven workflows connecting recruitment features with backend services.
 - Worked across authentication, job-management, and real-time workflow components supporting the recruitment platform.
- **100+ downloads | 4.5★**

CVMine | 2020 – Present | Recruitment & Job Platform | Backend Engineering

- Developed secure backend APIs supporting **CV encryption, job search, and employer-agent connections**.
 - Implemented API functionality supporting recruitment and candidate-management workflows.
 - Worked on secure handling of candidate-related data and backend integration.
 - Contributed to the platform's job-search and employer-agent interaction capabilities.
- **5.0★**

FileShore | 2023 – Present | Cloud Document Management Platform | Full-Stack Engineering

- Developed application and backend workflows for secure document management and sharing.
- Implemented secure-sharing functionality supporting controlled document access.
- Worked on **multi-factor authentication and access-control workflows**.
- Developed multi-level electronic-signature workflows for business documents.
- Contributed to secure document-management capabilities across application and backend layers.

TalliteSS | 2021 – Present | HR & RPO Management Platform | Backend Engineering

- Developed backend services supporting **candidate profiling and job-order management**.
- Implemented role-based access and workflow automation for staffing and RPO operations.
- Built API-driven workflows supporting candidate and job-order processes.
- Contributed to end-to-end HR process automation.

EDUCATION

B.Tech – Information Technology

Dr. Sivanthi Aditanar College of Engineering, Tiruchendur, Tamil Nadu | 2010 – 2013